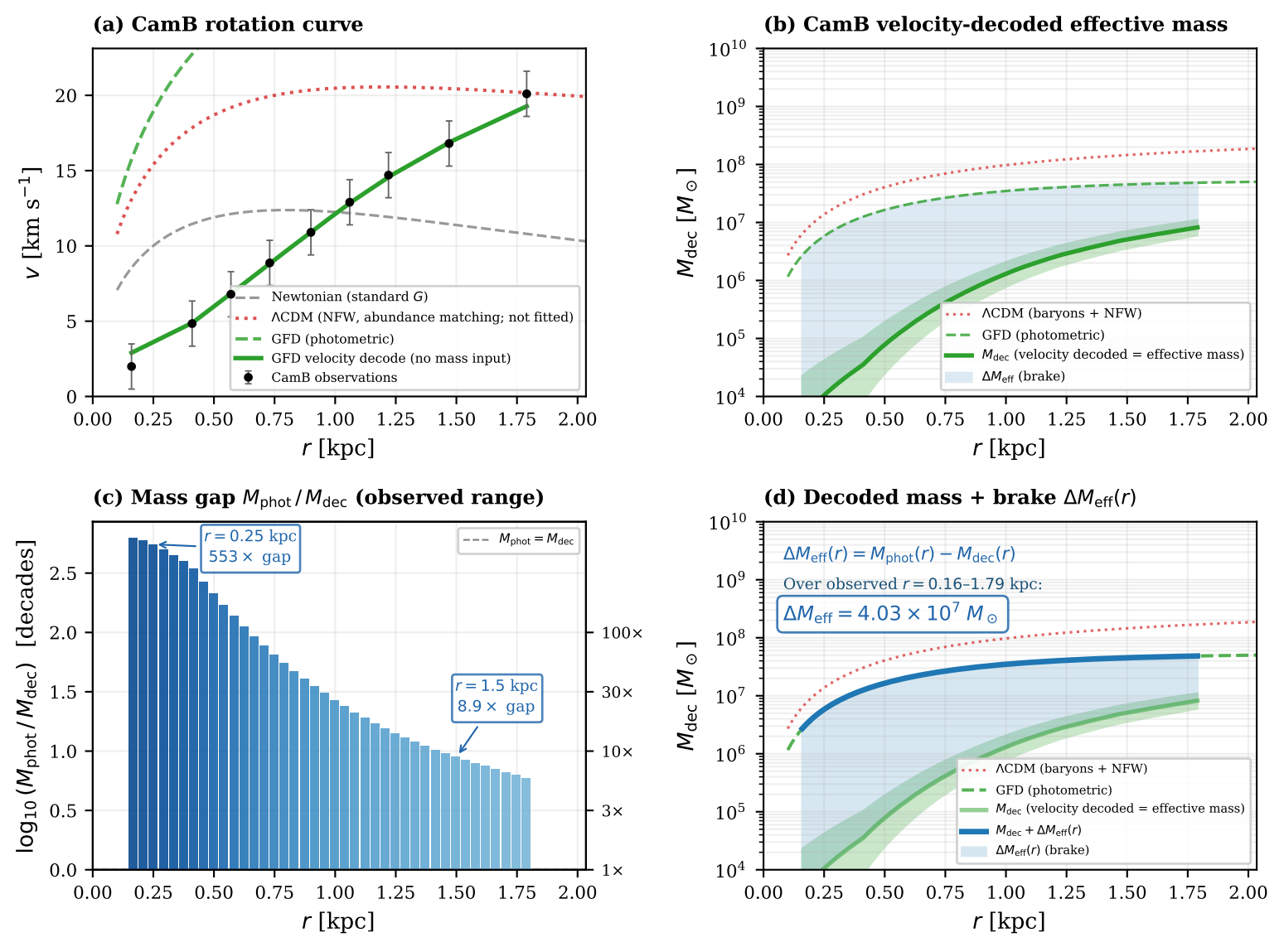

(a) CamB rotation curve
v [km s⁻¹]
r [kpc]
Newtonian (standard G)
ΛCDM (NFW, abundance matching; not fitted)
GFD (photometric)
GFD velocity decode (no mass input)
CamB observations
(b) CamB velocity-decoded effective mass
M_dec [M_⊙]
r [kpc]
ΛCDM (baryons + NFW)
GFD (photometric)
M_dec (velocity decoded = effective mass)
ΔM_eff (brake)
(c) Mass gap M_phot / M_dec (observed range)
log₁₀ (M_phot / M_dec) [decades]
r [kpc]
M_phot = M_dec
r = 0.25 kpc
553 × gap
r = 1.5 kpc
8.9 × gap
100×
30×
10×
3×
1×
(d) Decoded mass + brake ΔM_eff(r)
M_dec [M_⊙]
r [kpc]
ΔM_eff(r) = M_phot(r) − M_dec(r)
Over observed r = 0.16–1.79 kpc:
ΔM_eff = 4.03 × 10⁷ M_⊙
ΛCDM (baryons + NFW)
GFD (photometric)
M_dec (velocity decoded = effective mass)
M_dec + ΔM_eff(r)
ΔM_eff(r) (brake)